MAHI

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Professional Summary

Software Engineer pursuing MCA with a strong foundation in **C/C++, Python, Object-Oriented Programming (OOP), Cloud Computing and REST APIs**. Solid understanding of **Operating Systems** and multi-threading concepts, along with Agile development practices. Experienced in building full-stack applications, including **AI-integrated tools** (LLMs/FastAPI) and hardware-software pipelines (ESP32). Focused on problem-solving, debugging, and developing reliable software solutions.

Education

National Institute of Technology Raipur — **Aug 2024 – Present**
Master of Computer Applications (MCA) — *Raipur, India*

Bundelkhand University — **Jul 2021 – Jul 2024**
Bachelor of Science (B.Sc.) — *Jhansi, India*

Projects

AskLio AI Chatbot | ***React, Python (FastAPI), PostgreSQL, Groq/LLaMA AI*** — Live Demo |

- Engineered a ChatGPT-style AI application using **AI tools** (Groq LLaMA) and Python backend (FastAPI).
- Integrated Server-Sent Events (SSE) for real-time AI token streaming and managed persistent PostgreSQL chat history.
- Implemented secure JWT authentication and token-based session handling across REST API endpoints.

Tech: Python, FastAPI, React, Groq API (AI), PostgreSQL, JWT, Vercel

GitOrbit – GitHub Tracker | ***Next.js, FastAPI, MongoDB, GitHub REST API*** — Live Demo |

- Built a full-stack GitHub analytics dashboard to track followers, following, repository activity, and historical audience changes, deployed on Vercel (frontend) and Render (backend).
- Integrated GitHub REST API with token authentication to sync profile/repository data into MongoDB snapshots.
- Implemented snapshot comparison and append-only event logs to detect follow/unfollow events over time.
- Designed responsive dashboard UI with search, filters, and manual sync for real-time insights.

Tech: Next.js, FastAPI, MongoDB, GitHub REST API, Vercel

Wi-Fi CSI Human Activity Recognition | ***Python, ESP32, Multi-threading, NumPy***

- Designed a hardware-software sensing pipeline using ESP32 Wi-Fi CSI to recognize human activities via signal processing.
- Developed a real-time Python GUI logger utilizing **multi-threading** to handle concurrent data capturing, labeling, and CSV storage without blocking the main process.
- Applied OOP principles to structure the data parsing logic for scalable and reproducible dataset generation.

Tech: C++ (ESP32), Python, Multi-threading, NumPy, Pandas, Matplotlib

Technical Skills

Languages: C, C++, Python, Java, JavaScript, SQL
Core Concepts: Object-Oriented Programming (OOP), Operating Systems, Multi-threading, DSA, Agile
AI Tools & Tech: Groq API (LLaMA), Spring AI, AI Prompting, Machine Learning Concepts
Backend & DB: FastAPI, Pydantic, Node.js, REST APIs, PostgreSQL, MongoDB, MySQL
Cloud & DevOps: Linux, Docker, Kubernetes, Bash Scripting, Cloud Computing, GitHub Actions, CI/CD.
AI & Data Processing: NumPy, Pandas, Matplotlib, Scikit-learn
Tools & Frontend: Git, GitHub, Vercel, Render, React, Next.js, Tailwind CSS.

Awards & Achievements

- Solved **150+** data structures and algorithms problems utilizing **C++ and Python** on LeetCode and GeeksforGeeks.
- Successfully completed an intensive **C++ DSA Bootcamp** (MySirG), mastering Object-Oriented principles and pointers.
- Earned **Problem Solving** certification from HackerRank, demonstrating strong algorithmic thinking.
- Completed Full Stack Web Development training (iNeuron) in an Agile learning environment.